

Practical week-4

Step 1 — Create Database

The screenshot displays the SQL Developer interface. The left sidebar shows the 'SCHEMAS' tree with 'kldb', 'p_week1', 'p_week2', 'sweek1', and 'sys'. The main query editor contains two lines of SQL: 'create database pweek4;' and 'use pweek4;'. The right sidebar shows 'SQLAdditions' with a message: 'Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.' The bottom section shows the 'Output' window with a table of execution results.

#	Time	Action	Message	Duration / Fetch
1	14:31:10	create database pweek4	1 row(s) affected	0.016 sec
2	14:31:54	use pweek4	0 row(s) affected	0.000 sec

Step 2 — Create Books Table (DDL + Constraints)

The screenshot shows the SQL Developer interface with the following components:

- Navigator:** Shows a tree view of schemas including kldb, p_week1, p_week2, sweek1, and sys.
- Query Editor:** Contains the following SQL script:

```
1 • create database pweek4;
2 • use pweek4;
3 • CREATE TABLE Books
4 • (
5 •     book_id INT AUTO_INCREMENT PRIMARY KEY,
6 •
7 •     title VARCHAR(200) NOT NULL,
8 •
9 •     author VARCHAR(150) NOT NULL,
10 •
11 •     genre VARCHAR(100),
12 •
13 •     description TEXT,
14 •
15 •     published_year INT
16 • );
17
```
- SQLAdditions:** Displays a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."
- Output:** Shows the execution results of the SQL script:

#	Time	Action	Message	Duration / Fetch
1	14:31:10	create database pweek4	1 row(s) affected	0.016 sec
2	14:31:54	use pweek4	0 row(s) affected	0.000 sec
3	14:33:54	CREATE TABLE Books (book_id INT AUTO_INCREME...	0 row(s) affected	0.031 sec
- Schema:** p_week1
- Object Info:** Session

Step 3 — Describe Books Table

The screenshot shows the MySQL Workbench interface. The 'Query 1' tab contains the following SQL code:

```
CREATE TABLE Books (
  book_id INT AUTO_INCREMENT PRIMARY KEY,
  title VARCHAR(200) NOT NULL,
  author VARCHAR(150) NOT NULL,
  genre VARCHAR(100) YES,
  description text YES,
  published_year int YES
```

The 'Result Grid' shows the table structure:

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	<u>NONE</u>	auto_increment
title	varchar(200)	NO		<u>NONE</u>	
author	varchar(150)	NO		<u>NONE</u>	
genre	varchar(100)	YES		<u>NONE</u>	
description	text	YES		<u>NONE</u>	
published_year	int	YES		<u>NONE</u>	

The 'Output' pane shows the 'Action Output' table:

#	Time	Action	Message	Duration / Fetch
1	14:31:10	create database pweek4	1 row(s) affected	0.016 sec
2	14:31:54	use pweek4	0 row(s) affected	0.000 sec
3	14:33:54	CREATE TABLE Books (book_id INT AUTO_INCREM...	0 row(s) affected	0.031 sec
4	14:36:29	describe Books	6 row(s) returned	0.000 sec / 0.000 sec

A sidebar on the right contains a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."

Step 4 — Insert 10 Books and Display

The screenshot shows the MySQL Workbench interface. The 'Query 1' tab contains the following SQL code:

```
);
describe Books;
INSERT INTO Books (
  title,
  author,
  genre,
  description,
  published_year
)
VALUES (
  'The Martian',
  'Andy Weir',
  'Science Fiction',
  'An astronaut struggles to survive alone on Mars.',
  2011
),
(
  'The Martian',
  'Andy Weir',
  'Science Fiction',
  'An astronaut struggles to survive alone on Mars.',
  2011
),
(
  'The Martian',
  'Andy Weir',
  'Science Fiction',
  'An astronaut struggles to survive alone on Mars.',
  2011
),
(
  'The Martian',
  'Andy Weir',
  'Science Fiction',
  'An astronaut struggles to survive alone on Mars.',
  2011
),
(
  'The Martian',
  'Andy Weir',
  'Science Fiction',
  'An astronaut struggles to survive alone on Mars.',
  2011
),
(
  'The Martian',
  'Andy Weir',
  'Science Fiction',
  'An astronaut struggles to survive alone on Mars.',
  2011
),
(
  'The Martian',
  'Andy Weir',
  'Science Fiction',
  'An astronaut struggles to survive alone on Mars.',
  2011
),
(
  'The Martian',
  'Andy Weir',
  'Science Fiction',
  'An astronaut struggles to survive alone on Mars.',
  2011
),
(
  'The Martian',
  'Andy Weir',
  'Science Fiction',
  'An astronaut struggles to survive alone on Mars.',
  2011
),
(
  'The Martian',
  'Andy Weir',
  'Science Fiction',
  'An astronaut struggles to survive alone on Mars.',
  2011
);
```

The 'Output' pane shows the 'Action Output' table:

#	Time	Action	Message	Duration / Fetch
1	14:31:10	create database pweek4	1 row(s) affected	0.016 sec
2	14:31:54	use pweek4	0 row(s) affected	0.000 sec
3	14:33:54	CREATE TABLE Books (book_id INT AUTO_INCREM...	0 row(s) affected	0.031 sec
4	14:36:29	describe Books	6 row(s) returned	0.000 sec / 0.000 sec
5	14:39:56	INSERT INTO Books (title, author, genre, descri...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec

A sidebar on the right contains a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."

Step 5 — Add Embedding Column (JSON)

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```
97 • describe Books;
98 • SELECT
99     book_id,
100     title,
101     author,
102     genre,
103     published_year
104 FROM Books
105 ORDER BY book_id;
```

book_id	title	author	genre	published_year
1	The Martian	Andy Weir	Science Fiction	2011
2	Dune	Frank Herbert	Science Fiction	1965
3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	1979
4	Pride and Prejudice	Jane Austen	Romance	1813
5	The Great Gatsby	F. Scott Fitzgerald	Classic	1925
6	Foundation	Isaac Asimov	Science Fiction	1951
7	The Hobbit	J.R.R. Tolkien	Fantasy	1937
8	Enders Game	Orson Scott Card	Science Fiction	1985

#	Time	Action	Message	Duration / Fetch
2	14:31:54	use pweek4	0 row(s) affected	0.000 sec
3	14:33:54	CREATE TABLE Books (book_id INT AUTO_INCRE...	0 row(s) affected	0.031 sec
4	14:36:29	describe Books	6 row(s) returned	0.000 sec / 0.000 sec
5	14:39:56	INSERT INTO Books (title, author, genre, des...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec
6	14:40:58	describe Books	6 row(s) returned	0.000 sec / 0.000 sec
7	14:42:00	SELECT book_id, title, author, genre, publis...	10 row(s) returned	0.000 sec / 0.015 sec

Step 6 — Update Book Embeddings

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```
91     'Twenty Thousand Leagues Under the Seas',
92     'Jules Verne',
93     'Adventure',
94     'A fantastic underwater adventure aboard a mysterious submarine.',
95     1870
96 );
97 • describe Books;
98 • SELECT
99     book_id,
100     title,
101     author,
102     genre,
103     published_year
104 FROM Books
105 ORDER BY book_id;
106
107 • ALTER TABLE Books
108 ADD COLUMN embedding JSON;
```

#	Time	Action	Message	Duration / Fetch
3	14:33:54	CREATE TABLE Books (book_id INT AUTO_INCRE...	0 row(s) affected	0.031 sec
4	14:36:29	describe Books	6 row(s) returned	0.000 sec / 0.000 sec
5	14:39:56	INSERT INTO Books (title, author, genre, des...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec
6	14:40:58	describe Books	6 row(s) returned	0.000 sec / 0.000 sec
7	14:42:00	SELECT book_id, title, author, genre, publis...	10 row(s) returned	0.000 sec / 0.015 sec
8	14:43:00	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec

Step 7 — Display Book Embeddings

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```
122 WHERE book_id = 4;
123 UPDATE Books
124 SET embedding = JSON_ARRAY(0.15, 0.30, 0.80)
125 WHERE book_id = 5;
126 UPDATE Books
127 SET embedding = JSON_ARRAY(0.95, 0.65, 0.20)
128 WHERE book_id = 6;
129 UPDATE Books
130 SET embedding = JSON_ARRAY(0.30, 0.75, 0.90)
131 WHERE book_id = 7;
132 UPDATE Books
133 SET embedding = JSON_ARRAY(0.88, 0.85, 0.15)
134 WHERE book_id = 8;
135 UPDATE Books
136 SET embedding = JSON_ARRAY(0.20, 0.55, 0.95)
137 WHERE book_id = 9;
138 UPDATE Books
139 SET embedding = JSON_ARRAY(0.70, 0.90, 0.40)
140 WHERE book_id = 10;
```

#	Time	Action	Message	Duration / Fetch
15	14:46:59	UPDATE Books SET embedding = JSON_ARRAY(0.15, ...	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.000 sec
16	14:47:16	UPDATE Books SET embedding = JSON_ARRAY(0.95, ...	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.000 sec
17	14:47:30	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90)	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.000 sec
18	14:47:44	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15)	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.015 sec
19	14:47:57	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95)	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.000 sec
20	14:48:16	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40)	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.000 sec

Query Completed

Step 8 — L2 Distance Query

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```
140 WHERE book_id = 10;
141
142 SELECT
143     book_id,
144     title,
145     embedding
146 FROM Books
147 ORDER BY book_id;
148
```

book_id	title	embedding
1	The Martian	[0.90, 0.80, 0.20]
2	Dune	[0.85, 0.75, 0.25]
3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
4	Pride and Prejudice	[0.10, 0.20, 0.90]
5	The Great Gatsby	[0.15, 0.30, 0.80]
6	Foundation	[0.95, 0.65, 0.20]
7	The Hobbit	[0.30, 0.75, 0.90]
8	Enders Game	[0.88, 0.85, 0.15]

#	Time	Action	Message	Duration / Fetch
16	14:47:16	UPDATE Books SET embedding = JSON_ARRAY(0.95, ...	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.000 sec
17	14:47:30	UPDATE Books SET embedding = JSON_ARRAY(0.30, ...	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.000 sec
18	14:47:44	UPDATE Books SET embedding = JSON_ARRAY(0.88, ...	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.015 sec
19	14:47:57	UPDATE Books SET embedding = JSON_ARRAY(0.20, ...	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.000 sec
20	14:48:16	UPDATE Books SET embedding = JSON_ARRAY(0.70, ...	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.000 sec
21	14:48:58	SELECT book_id, title, embedding FROM Books	10 row(s) returned	0.000 sec / 0.000 sec

Query Completed

Step 9 — Cosine Similarity Query

The screenshot shows a database IDE interface. The left sidebar displays a schema tree with 'p_week1' selected. The main query editor shows a SQL query for Step 9, which calculates the cosine similarity between book embeddings. The query is as follows:

```
SELECT
  book_id,
  title,
  author,
  genre,
  ROUND(
    SQRT(
      POW(

```

The 'Result Grid' shows the following data:

book_id	title	author	genre	l2_distance
1	The Martian	Andy Weir	Science Fiction	0
8	Enders Game	Orson Scott Card	Science Fiction	0.0735
2	Dune	Frank Herbert	Science Fiction	0.0866

The 'Output' pane shows the execution log for 'Result 6', including update and select statements with their respective durations and row counts.

Step 10 — Final Similarity Result

The screenshot shows the same database IDE interface for Step 10. The query editor displays a more complex SQL query that calculates the cosine similarity using JSON functions. The query is as follows:

```
genre,
ROUND(
(
CAST(
JSON_EXTRACT(embedding, '$[0]')
AS DECIMAL(10,6)
) * 0.90
+
CAST(

```

The 'Result Grid' shows the final similarity results:

book_id	title	author	genre	cosine_similarity
1	The Martian	Andy Weir	Science Fiction	1
2	Dune	Frank Herbert	Science Fiction	0.9986
8	Enders Game	Orson Scott Card	Science Fiction	0.9983

The 'Output' pane shows the execution log for 'Result 7', including update and select statements.